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LESSON PLAN	**COURSE: MHF4U**
Date: Lesson #14	**Unit of Study: Chapter #8 Combining Functions**
Materials / Resources: - **Advanced Function 12 Study Guide** - **Chapter #8 Pre-assessment Quiz** - **Chapter #8 Formula, Aid and Homework Sheet(s)**	**Lesson Focus** - **Sum and Difference of Functions** - **Product and Quotients of Functions** - **Composite Functions** - **Inequalities of Combined Functions**

Prior Knowledge:

- The use of '$f(x)$' and the input and output of a function
- Simple effects on functions when adding or subtracting a constant to a function

Overall Expectations:

1. Determine functions that result from the addition, subtraction, multiplication, and division of two functions and from the composition of two functions, describe some properties of the resulting functions, and solve related problem

Specific Expectations:

2.1 Determine, through investigation using graphing technology, key features (e.g., domain, range, maximum/minimum points, number of zeros) of the graphs of functions created by adding, subtracting, multiplying, or dividing functions [e.g., $f(x) = 2^{-x}$ sin $sin\ 4x$, $g(x) = x^2 + 2^x$, $h(x) = \frac{\sin sin\ x}{\cos cos\ x}$], and describe factors that affect these properties

2.2 Recognize real-world applications of combinations of functions (e.g., the motion of a damped pendulum can be represented by a function that is the product of a trigonometric function and an exponential function; the frequencies of tones associated with the numbers on a telephone involve the addition of two trigonometric functions), and solve related problems graphically

2.3 Determine, through investigation, and explain some properties (i.e., odd, even, or neither; increasing/decreasing behaviours) of functions formed by adding, subtracting, multiplying, and dividing general functions [e.g., $f(x) + g(x)$, $f(x)g(x)$]

2.4 Determine the composition of two functions [i.e., $f(g(x))$] numerically (i.e., by using a table of values) and graphically, with technology, for functions represented in a variety of ways (e.g., function machines, graphs, equations), and interpret the composition of two functions in real-world applications

2.5 determine algebraically the composition of two functions [i.e., $f(g(x))$], verify that $f(g(x))$is not always equal to $g(f(x))$ [e.g., by determining $f(g(x))$ and $g(f(x))$, given $f(x) = x + 1$ and $g(x) = 2x$], and state the domain [i.e., by defining $f(g(x))$ for those x-values for which $g(x)$ is defined and for which it is included in the domain of $f(x)$] and the range of the composition of two functions

2.6 Solve problems involving the composition of two functions, including problems arising from real-world applications

2.7 Demonstrate, by giving examples for functions represented in a variety of ways (e.g., function machines, graphs, equations), the property that the composition of a function and its inverse function maps a number onto itself [i.e., $f^{-1}(f(x)) = x$ and $f\left(f^{-1}(x)\right) = x$ demonstrate that the inverse function is the reverse process of the original function and that it undoes what the function does]

2.8 Make connections, through investigation using technology, between transformations (i.e., vertical and horizontal translations; reflections in the axes; vertical and horizontal stretches and compressions to and from the x- and y-axes) of simple functions $f(x)$ [e.g., $f(x) = x^3 + 20$, $f(x) = \sin sin\, x$, $f(x) = \log log\, x$] and the composition of these functions with a linear function of the form $g(x) = A(x + B)$

Learning Goals:

- Students will be able to add or subtract two or more functions graphically using the superposition principal
- Students will be able to add or subtract two or more functions algebraically
- Students will know how to combine and simplify a product of functions algebraically
- Students will know how to combine and simply a quotient of functions algebraically, and identify any restrictions on the variable
- Students will have knowledge on how to graph the sum, difference, product or quotient of functions with and without graphing technology and identify the key characteristics of the graph
- Students will be able to determine algebraically the composition of two or more functions
- Students will understand the effect of operating on a variable by a function followed by its inverse
- Students will know how to solve inequalities of combined functions algebraically
- Students will know how to solve inequalities of combined functions graphically, with and without technology, using a variety of strategies
- Students will be able to answer problems involving various combination of functions

Success Criteria:

1. By the end of this lesson I will be able to add or subtract two or more functions graphically using the superposition principal
2. By the end of this lesson I will be able to add or subtract two or more functions algebraically
3. By the end of this lesson I will be able to combine and simplify a product of functions algebraically
4. By the end of this lesson I will be able to combine and simplify a quotient of functions algebraically, and identify any restrictions on the variable
5. By the end of this lesson I will be able to graph the sum, difference, product, or quotient of functions with and without graphing technology and identify the key characteristics of the graph
6. By the end of this lesson I will be able to algebraically determine the composition of two or more functions
7. By the end of this lesson I will be able to understand the effect of operating on a variable by a function followed by its inverse
8. By the end of this lesson I will be able to solve inequalities of combined functions algebraically
9. By the end of this lesson I will be able to solve inequalities of combined functions graphically, with and without technology, using a variety of strategies
10. By the end of this lesson I will be able to solve problems involving various combination of functions

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<tr><td colspan="2"></td></tr>
<tr><td colspan="2">Assessment Strategies:

<u>Diagnostic</u>: chapter #8 pre-assessment quiz

<u>Formative</u>: homework for chapters 8.1, 8.2, 8.3 & 8.4

<u>Summative</u>: test #5, assignment #2</td></tr>
<tr><td>Time:

Part 1</td><td>Introduction:

Students will be given a pre-assessment quiz to determine their knowledge standings for combined functions, and then proceed from their answers</td></tr>
<tr><td>Time:

Part 2</td><td>Lesson:

<u>Chapter 8.1 Sum and Difference of Functions</u>
Start by introducing the superposition principle, which states that the sum of two or more functions can be found by adding the y-coordinates of the function at each x-coordinate. Give a simple example of how this principle would be put to play (e.g. the sum of $f(x) = 2x$ and $g(x) = 2$ is $f(x)$ going up 2 units for every x-coordinate). Also establish that the superposition principle also applies to differences of two functions.

<u>Chapter 8.2 Product and Quotients of Functions</u>
Introduce the notation for a combined function that represents the product and quotient of two functions. Investigate the domain and range of the combined functions and how they would change from the original functions. Take notice on the domain of the quotient of two functions as the denominator cannot equal to zero.

<u>Chapter 8.3 Composite Functions</u>
Introduce the notations for composite functions as they could be written as $f(g(x))$ or $f \circ g(x)$, which defines the same function. For the instructions of a composite function, explain that a function $g(x)$ would replace all x's within the function $f(x)$ and therefore it is the input of the $f(x)$ function. To evaluate a composite function, either first evaluate it for $g(x)$ first then substitute in $f(x)$ or simplify the function $f(g(x))$ then proceed from the answer. Take note of functions such as $f(f(x))$ and $f^{-1}g(x))$, where the former is simply inputting a function to itself and the latter is done by first finding out the inverse of $f(x)$ and then inputting $g(x)$ into the inverse.

<u>Chapter 8.4 Inequalities of Combined Functions</u>
Taking notes from previous chapters, an inequality of combined functions is solved by graphing the two functions within the same graph and then proceeding with observing the range of the domain where one functions is either greater or lesser than the other. Using different functions as examples (sin $sin\ x$, cos $cos\ x$, log $log\ x$, $\frac{f(x)}{g(x)}$) to solve for inequalities, again putting them on the same axes, determine the areas (thus the domain) where one function is greater or lesser than the other.</td></tr>
<tr><td>Time:

Part 3</td><td>Wrap-up:</td></tr>
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	Ask students to demonstrate their acquired knowledge by applying them to example problems and then answer any unresolved questions. Use KWL chart to better understand student's learning needs for chapter #8.

Notes, reminders & homework:

<u>Reminder</u>:

- Test #5 (to be finished after lesson #14)
- Students should be receiving their assignment #2 after current class. Students have one week from current date to finish the assignment.

<u>Homework</u>:

Chapter 8.1: # 1 – 8, 10, 13, 17
Chapter 8.2: # 1 – 10
Chapter 8.3: # 1 – 3, 5, 7 – 10, 13, 15
Chapter 8.4: # 1 – 3, 6 – 8, 10, 11

Lesson Reflection: